

Literature position for the threshold-entry contest

Survey and verified bibliography

1 September 2026

1 Purpose and verification standard

This document settles the literature check flagged as the gating open item in `PROOFS.tex` (§Open items, item 4) and in the referee handover (R1 Comment 4). The question it answers is narrow: is the mechanism claimed as novel — an identical nominal entry cost c made asymmetric in utility by concavity, so that *who* enters a fixed-prize contest is determined by the wealth distribution rather than by ability — already occupied?

Every entry in `refs.bib` has author list, title, journal, year, volume and pages verified against at least one of: the article’s own title page; the publisher’s article page; or the reference list of a published peer-reviewed article. Where a claim below describes what a paper *does* rather than what it is, the evidence level is marked:

[F] full text read;

[A] abstract or publisher summary only — model details are reported as the abstract states them and have not been checked against the body;

[C] characterised via a citing paper’s description, not the source itself.

Where the text read was a working paper or draft rather than the published article, the tag says so — “[F, WP version]” or “[F, draft version]” — and proposition numbers quoted from it may not match the published version.

Claims marked [A] or [C] are not safe to cite as statements about what a paper proves. They are safe as statements about what a paper is about. Anything that enters `PROOFS.tex` as a positioning claim should be upgraded to [F] first.

2 The closest paper on the affordability channel, read in full

Schroyen and Treich (2016) is the nearest occupant of the mechanism, and the one against which the contribution has to be stated. It has now been read in full [F]. The version read is TSE Working Paper 16-699 dated 11 April 2016, which the published *Games and Economic Behavior* article replaced; the published text has not been checked line by line against it, so any quotation of a theorem number should be confirmed there before submission.

Correction to the previous draft of this document. An earlier version of this survey, written from the abstract, stated that their setting produces no comparative statics on the wealth distribution. That is wrong. Their Theorem 3 signs the effect of a mean-preserving spread in wealth in the privilege contest. The overlap with P9 is therefore real and is set out below.

Primitives. A static, one-shot, two-player game. Appendix A.1 states that the only source of heterogeneity is wealth w_i . Three models are distinguished by whether the rent r and the effort x_i are commensurable with wealth inside u :

$$\begin{aligned} \text{privilege: } U_i &= u(w_i - x_i) + \pi_i r, \\ \text{ability: } U_i &= \pi_i u(w_i + r) + (1 - \pi_i)u(w_i) - c(x_i), \\ \text{rent-seeking: } U_i &= \pi_i u(w_i + r - x_i) + (1 - \pi_i)u(w_i - x_i), \end{aligned}$$

with π the power-logistic Tullock CSF of decisiveness m . Effort is continuous on a convex feasible set. Dynamic effects are listed in their conclusion as an extension not undertaken, as is an arbitrary number of players.

Consequence 1: the wealth/income distinction does no work. Because the model is static with no saving, w_i is a one-period resource level out of which effort is paid. That is formally the same object as the resource variable in `PROOFS.tex`. Describing our primitive as income and theirs as wealth is a labelling difference between two papers, not a difference between two models, and it cannot be used to separate them. The separation has to come from structure.

Consequence 2: our model is a privilege contest, confirmed from the functional form. Entry cost c is paid out of resources and V is separable and exogenous, so only the marginal-cost channel is live and the marginal-benefit channel is switched off by construction. This is now verified against $U_i = u(w_i - x_i) + \pi_i r$ rather than inferred from the taxonomy's name. It also fixes the open design question from the rebuild handover: making V money added to the resource would move the model to their rent-seeking contest, where the two wealth effects oppose and, under CARA, exactly cancel.

Consequence 3: the P9 overlap, stated honestly. Their Theorem 3 gives, for the symmetric privilege contest, that a small mean-preserving spread in wealth raises aggregate effort if and only if

$$2A(1 - m^2) > P, \quad A = -\frac{u''}{u'}, \quad P = -\frac{u'''}{u''},$$

reducing to $m < 2^{-1/2} \approx 0.707$ under CARA, to $m < 1$ under quadratic utility, and to $\gamma(\frac{1}{2} - m^2) > \frac{1}{2}$ under CRRA with coefficient γ . So a mean-preserving spread in the resource distribution, inside a privilege contest, is already in print.

Consequence 4: what survives, and why it is the contribution. Their result is local and heavily conditioned. It is a second-order effect of a small spread evaluated at a *symmetric* two-player equilibrium, obtained as a quadratic form in the Hessian of the best-response function (their Theorem 2), and its sign turns on the CSF decisiveness parameter and on the third derivative of u . Their own summary concedes that the wealth-inequality effects are the most complex in the paper and that signing a mean-preserving spread is very difficult without further restricting functional forms. The condition $u \in \Omega$ they need in the rent-seeking case — that a mean-preserving spread of background risk raises absolute risk aversion — is, by Eeckhoudt et al. (1996), a restriction on higher-order risk attitudes, and is stronger than DARA, being related to the risk vulnerability of Gollier and Pratt (1996).

P9-gen requires none of this. Its sign rule needs only that κ is strictly decreasing and that T is a pivot-spread, with the flip located at $w_{(k^*)}$ against the pivot x_0 . No third derivative, no prudence condition, no decisiveness parameter, no functional-form restriction, and the statement is global rather than a local second-order expansion. The contribution sentence should therefore be:

Moving the choice from the intensive to the extensive margin makes the mean-preserving-spread comparative static sign-determinate without higher-order restrictions on utility.

That is a sharp claim stated directly against a named published result, and it is stronger than anything the mechanism alone supports.

Consequence 5: a caveat that narrows the noise argument. They note, citing Jia (2008), that the power-logistic Tullock CSF can be derived as a rank-order tournament in which the noise terms are inverse-exponential, and remark that their contests may therefore also be read as rank-order tournaments. The deterministic-versus-noisy contrast drawn against Hopkins and Kornienko (2010) in §3 therefore does *not* extend to Schroyen–Treich. The remaining difference is narrower but real: their noise is one specific family, whereas F and G here are arbitrary subject only to the stated regularity.

Consequence 6: no entry margin. There is no participation decision anywhere in the paper. The zero-effort discontinuity in the best-response functions is a corner of the continuous effort choice, not a fixed-cost entry decision. The extensive margin is genuinely absent, which is what P5 and P7 require and what makes the contribution sentence above available.

A connection worth checking. They contrast their mean-preserving-spread results with the neutrality result of Bergstrom et al. (1986) for private provision of public goods, and note in a footnote to their Theorem 2 that the quadratic form vanishes under the Bergstrom–Varian conditions under which the distribution of agent characteristics has no effect on aggregate effort. Our anonymity proposition — that Δ depends on the entrant count alone and is invariant to the resource distribution — is a neutrality statement of the same shape, and it is precisely what makes P9 tractable. Whether the two are formally related, or merely rhyme, is not established here and should not be asserted without proof.

3 Hopkins and Kornienko: the standard result, read in full

Hopkins and Kornienko (2010) is the paper Referee 2 invoked when predicting that the standard result — inequality reduces status expenditure — would survive in this setting. It has now been read in full [F], and the reading changes the positioning in four specific ways.

Primitives. A continuum of contestants with endowment z drawn from G , private information. Each divides z between visible performance x and private consumption $y = z - x$. A continuum of indivisible rewards with values distributed H is assigned assortatively by rank in performance, so that in a separating equilibrium $G(z) = F(x) = H(s) = r$. Utility is $U(x, y, s)$ with monotonicity, complementarity and own-concavity conditions. The unique symmetric separating equilibrium solves a differential equation with the non-competitive optimum as boundary condition, and equilibrium performance strictly exceeds the cooperative level everywhere above rank zero, so the equilibrium is Pareto dominated from the contestants’ point of view. Existence and uniqueness are obtained by fitting the model into Mailath (1987).

Point 1: there is no extensive margin. Every contestant competes. The decision is how much performance to buy, never whether to compete at all. No participation constraint, no entry cost, no fixed fee. The binary entry decision that generates k^* in P5 has no counterpart in the paper. This is the cleanest available statement of what the rebuilt model adds and it is now verified rather than inferred.

Point 2: the contest is perfectly discriminating. The authors note explicitly that much of the surrounding contest literature assumes a partly stochastic award mechanism, and describe their own model by contrast as a perfectly discriminating rank-order tournament. The rebuilt model is on the stochastic side: scores are $\mu r_i + (1 - \mu)s_i e_i$ and the induced CDFs F and G carry all the content. The two models sit in different branches, and P-MU — where the sign of $\partial\Delta(0)/\partial Q$ turns on a hump-shaped weighting — is an object that cannot arise without noise.

Point 3: the wealth channel is a budget, not a fee. Endowment enters as the resource divided between performance and consumption. It sets how much performance a contestant can afford at the margin. It does not act through the utility cost of a fixed payment, which is the entire content of $\kappa(w) = u(w) - u(w - c)$ in the rebuild. These are different mechanisms and should not be described as the same one.

Point 4 — the most useful finding: the right stochastic order is already named. The comparative statics are conducted in the *dispersive order* and the *star order*, not in mean-preserving spreads. Their Definition 1, taken from Shaked and Shanthikumar (2007), is that F is less dispersed than G , written $F \leq_d G$, whenever $G^{-1}(r) - F^{-1}(r)$ is weakly increasing on $(0, 1)$; equivalently, at a fixed rank the more dispersed distribution is the less dense one. They also record that there is no clear relation between the dispersive order and second-order stochastic dominance, because SOSD compares distributions on both mean and dispersion while the dispersive order concerns dispersion alone.

This bears on open item 2 of PROOFS.tex, which observes that Theorem P9gen requires T to be a pivot-spread and that a general Rothschild–Stiglitz spread need not be monotone, leaving the general case open. The dispersive order is exactly the tool that literature uses for this difficulty, and the two conditions are close. Writing $T = \Lambda_\beta^{-1} \circ \Lambda_\alpha$, the dispersive order says $T(w) - w$ is increasing; if in addition $T(w) - w$ changes sign, it does so once, at a point x_0 , giving $T(w) \geq w$ above x_0 and $T(w) \leq w$ below it — which is the pivot-spread hypothesis. So the pivot-spread class is not an ad hoc restriction invented to make the proof work; it is, up to the crossing condition, the standard dispersion order of this literature. That is a considerably stronger position than the current text claims for itself, and it should be checked formally and then stated.

Direction of their result, stated carefully. Under rank indexing, their Proposition 4 assumes the minimum endowment is higher ex post, endowments are less dispersed ex post, and the maximum endowment is lower ex post. The conclusion is that performance is *higher* ex post on $[0, \hat{r}]$, where $\hat{r}$ is the single crossing point of the two endowment functions, and that utility rises at the bottom but falls for the middle and top. So less dispersion raises competition at the bottom and middle. Read in the other direction, more dispersion of endowments lowers performance there — which is the “standard result” Referee 2 had in mind, and it does hold.

Two cautions follow for P9. First, their conclusion is stated on an interval below a crossing point, not globally, so it is not a uniform-sign aggregate claim and should not be described as one. Second, their model already contains a crossing point at which effects change sign, though it is a crossing in the *endowment functions* with a corresponding sign change in *utility*, not a sign change in the direction of the expenditure response. P9’s pivot rule is a different object: a sign reversal in $dk^*/d(\text{spread})$ located at the pivot x_0 relative to the marginal entrant’s wealth $w_{(k^*)}$. The claim in PROOFS.tex §11 that the result “reconciles” the standard result with the motivating pattern is defensible only if stated at that level of precision. As drafted it is too loose, because it invites the reading that the standard result is uniform in sign, which the source does not say.

Genealogy of the dispersive order, corrected. Hopkins and Kornienko (2004) and Hopkins and Kornienko (2009) have now been read in full [F], and together they change what was said about the dispersive order in an earlier draft of this section. The claim there was that the order traces to the general stochastic-orders textbook of Shaked and Shanthikumar (2007) and merely happens to fit this problem. That understates it. The dispersive order, applied to exactly this class of problem, is Hopkins and Kornienko’s own methodological innovation, developed across these two papers before the 2010 paper uses it.

Hopkins and Kornienko (2004) is the earlier, income-indexing treatment: agents are indexed by income z itself, and equilibrium strategies and utilities are written as functions of z . Comparative statics there use first- and second-order stochastic dominance, together with two likelihood-ratio refinements — the monotone likelihood ratio (MLR) order and, for spreads, the unimodal likelihood ratio (ULR) order. Under the ULR order the paper’s Proposition 4 partitions the population into three zones by two threshold points $\hat{z}_-, \hat{z}_+$ — poor, middle, and rich — and signs the effect of a spread separately in each zone, with the middle zone (where the density ratio is highest) unambiguously seeing higher conspicuous consumption. This is a *two*-crossing-point result, more general than a single pivot.

Hopkins and Kornienko (2009) is explicit, in its own introduction, about why a second methodology was needed: the income-indexing approach of the 2004 paper cannot compare distributions with different supports, so it cannot handle a policy that raises the floor of the income distribution while taxing the top. The paper switches to indexing agents by *rank* $r = G(z)$ instead, and introduces the dispersive order for this purpose, citing Shaked (1982) directly (full citation added to `refs.bib` below) — rank-indexing plus the dispersive order is presented there as a new technique built specifically to sign comparative statics of a competitive equilibrium under a change in the underlying distribution, which is precisely the class of problem P9-gen belongs to.

Structurally, Hopkins and Kornienko (2009) Propositions 4 and 5 are close cousins of P9-gen. Proposition 4 gives a sufficient condition for a full Pareto improvement: the poorest agent no worse off, together with the distribution becoming *more* dispersed in their sense, is enough for utility to rise almost everywhere. Proposition 5 is the mirror image: the poorest richer, the richest poorer, and the distribution *less* dispersed, then utility falls for the middle and upper ranks — this is the source proposition the 2010 paper’s Proposition 4 (discussed above in §3) generalises. In both cases the sign of the welfare effect is pinned by a single crossing point $\hat{r}$ of the two distribution functions, in exactly the way P9-gen pins the sign of the entry-count effect by the pivot x_0 . The two constructions are not the same theorem — theirs crosses in the distribution functions and signs a welfare comparison, P9-gen crosses in $T(w) - w$ and signs an entry-count comparison — but they are the same technique applied to structurally analogous problems, and P9-gen should cite this lineage rather than only the general textbook.

A concrete open question this raises. Hopkins and Kornienko (2004)’s ULR result permits *two* crossing points and three sign regions, not one. P9-gen’s pivot-spread hypothesis is a single-crossing object. Whether a two-pivot generalisation of P9-gen exists — for spreads that compress agents in the middle of the wealth distribution while still expanding the tails, which a single pivot cannot represent — is an open question raised directly by this reading and is not addressed anywhere in the current bundle.

Consequence for the income/wealth question. Both papers work entirely in *income* space: the primitive is z , described throughout as income, divided in a single period between a positional good and a non-positional good, with no saving and no mention of wealth as a stock. This is independent confirmation, from the second major reference paper in this literature (after

Schroyen–Treich), of the point made when the primitive in `PROOFS.tex` was discussed: a one-period resource divided between an entry cost and residual consumption is standard practice as “income” in this literature, not “wealth,” and the field’s own vocabulary supports the relabelling already adopted.

No entry margin, confirmed again. As with Hopkins–Kornienko (2010) and Schroyen–Treich, there is no participation decision in either paper: $x \in [0, z/p]$ is a continuous choice on a convex feasible set. The extensive-margin gap the entry_contest bundle fills is confirmed a third time.

Hopkins (2012) and Hopkins (2008) are adjacent, not yet read in full. [C]

4 Endogenous entry into contests, read in full

Four papers in this strand have now been read in full [F]: Levin and Smith (1994) and Morgan et al. (2012) in their published versions, Fu et al. (2015) in its published version, and Fu and Lu (2010) as MPRA Paper 945 (November 2006), the working version of the *Economic Inquiry* article. Together they settle the entry-structure question more sharply than the abstracts allowed.

The identity problem, stated by the literature itself. All four papers assume identical potential entrants. Every one of them then confronts the same difficulty: with identical agents, the pure-strategy count is determined but the identity of who enters is not. Levin and Smith (1994), p. 586 (main text; footnote 2 is attached to the preceding sentence and concerns Engelbrecht-Wiggans), say of the prior literature’s deterministic asymmetric equilibria that the process by which potential bidders divide into entrants and non-entrants is not explained. Morgan et al. (2012), p. 441, say in their own model that because all players are identical the identity of those opting in is not uniquely determined. The literature has three devices for this. Levin and Smith restore symmetry by mixed entry, so the number of entrants becomes stochastic (binomial). Fu and Lu (2010) (footnote 6) and Morgan et al. (2012) (Proposition 1) use sequential entry with observability, so arrival order breaks the tie; Morgan et al. add private random delays to make the order well defined. Fu et al. (2015) take the mixed route and, because entrants do not observe the number of rivals, need Dasgupta–Maskin’s theorem for two-dimensional discontinuous games to establish existence (their Theorem 1).

P5 is the fourth device, and it differs from the other three in kind: the resource ordering $\kappa(w_{(1)}) < \kappa(w_{(2)}) < \dots$ is a property of the *agents*, not of a timing protocol or a randomisation, so the pure-strategy count k^* is deterministic under *simultaneous* entry with no mixing and no arrival order — and, by the count-invariance proposition in `PROOFS.tex`, the same in every pure-strategy equilibrium.

The claim should stop there, and an earlier version of this survey did not. Heterogeneous costs do not by themselves pin *which* agents enter: with $\Delta = (12, 10, 1)$ and $\kappa = (2, 5, 8)$ the sets $\{1, 2\}$, $\{1, 3\}$ and $\{2, 3\}$ are all equilibria, one of them with the richest challenger absent. What is delivered is the count, plus existence of the assortative equilibrium, plus its uniqueness under a stated extra condition; absent that condition, assortativity is a selection. That is still a real advance on mixing, arrival order and random delays, each of which leaves even the assortative outcome undefined — but it is a smaller claim than “identity is a property of the agents”, and the two footnotes cited above are the places to hang the smaller one.

Levin–Smith. N identical risk-neutral potential bidders, fixed entry cost c paid ex ante before learning one’s value (contrast Samuelson’s interim costs), n revealed before bidding. Their n^* is

defined as the largest integer at which the marginal entrant’s expected gain is non-negative — the same object as k^* for identical agents — and their symmetric equilibrium is in mixed strategies with $n \sim \text{Binomial}(N, q^*)$. Two results bear on the open welfare item (R2 Comment 3). In common-value auctions free entry is *excessive* by a business-stealing argument (Proposition 3, citing Mankiw and Whinston (1986)), and the seller taxes entry; in independent-private-value auctions free entry is *optimal* because the marginal entrant’s private gain equals the social gain, via $V_n - W_n = n(V_n - V_{n-1})$ (Proposition 6, equation 18).

Which branch applies is settled by a *criterion*, not by resemblance: Proposition 6 holds exactly when equation (18) holds, and (18) requires the marginal entrant’s private gain to equal the increment to total value. With a prize fixed at V we have $V_n \equiv V$, so $V_n - V_{n-1} = 0$: the social gain from any entrant beyond the first is *exactly zero* while the social cost is $c > 0$. The paper states this directly (p. 596): “In CV auctions, social gains are zero (and therefore smaller than social costs) for all $n \geq 2$.” The entry contest model is therefore in the common-value branch *by construction* — V exogenous and fixed is $V_n \equiv V$ — and the benchmark is Proposition 3, with the expected verdict of excessive entry. Only a prize that grew with participation could reinstate Proposition 6.

Their Proposition 9 adds that welfare falls monotonically as the pool N grows beyond n^* , through the coordination cost of mixed entry; Proposition 8 is the supporting statement that the probability of *no* entry rises with N under an optimal mechanism. That is a different mechanism from P7’s cap, which is a count bound with no coordination content, but both say that a larger pool does not help beyond a point.

Their equation (9), $(1 - q)^{N-1}V = c$, requires care. It is *not* an entry-equilibrium condition: it is the first-order condition of the welfare problem, obtained by differentiating their equation (8), $S = [1 - (1 - q)^N]V - qNc$, and setting $\partial S / \partial q = 0$. It characterises the socially optimal entry probability, which the seller’s optimal entry fee e^* is chosen to induce; the paper’s own proof of Proposition 8 writes it as $(1 - q_N^s)^{N-1} = c/V$, with s for social. The bidders’ equilibrium condition is their equation (6), $B_i(q^*, \Omega) = 0$. Equation (9) is nonetheless algebraically identical to Fu et al. (2015)’s Definition 1, which reaches the same equation from the opposite direction — as a *lower bound* on equilibrium entry, derived from the win-only-if-alone deviation payoff. The coincidence is worth stating, but the two must not be identified: in common-value auctions the free-entry equilibrium lies *above* q^s (Proposition 3), and reading (9) as an equilibrium condition erases the very wedge that Proposition 3 is about.

Fu–Jiao–Lu, and the accounting behind P7. $M \geq 2$ identical risk-neutral potential bidders, common fixed entry cost, generalised nested lottery contest, prize purse capped at V and allowed to be contingent on the realised number of entrants. Their equation (2) is a rent-accounting inequality: expected prize mass paid out is at least expected entry costs plus expected bidding costs, so

$$[1 - (1 - q)^M] V \geq Mq(\Delta + \mathbb{E}[x^\alpha]),$$

which implies that the expected number of entrants Mq cannot exceed V/Δ , since the bracket is at most 1 and $\mathbb{E}[x^\alpha] \geq 0$. Cite *Definition 2 alongside equation (2)* for this: Fu et al. never display the count bound, and what they derive from (2) is a chain about *bids* — (3), then (4) by convexity, (5), the bid bound (6), and Theorem 2’s single-peakedness. The count bound is instead the content of their Definition 2, $\bar{q} = \arg \max\{q \in [0, 1] : (1 - (1 - q)^M)V \geq Mq\Delta\}$, every feasible point of which satisfies $Mq\Delta \leq V$. A referee sent to equation (2) alone will find only the bid chain and conclude the attribution is loose. That is the same accounting logic as P7. P7’s content is therefore not a new inequality but the deterministic, heterogeneous, utility-unit version of a known one: it bounds a pure-strategy count rather than an expected count, it uses

κ in utility units rather than a nominal cost, and it holds under fixed rules with no designer. The paper should say exactly that. What Fu et al. build on top — the uniform upper bound on expected overall *bids* (Theorem 2), its single-peakedness in q , its implementation by Tullock contests with entry fees (Theorem 7), and the optimal shortlisting rule (Lemma 6, Theorem 9, with $\bar{M} = \min\{N : V/N < \alpha\Delta/(\alpha - 1)\}$) — are design results with an effort-maximising organiser and have no counterpart in the entry contest model, which has no designer and a single fixed prize. Their Theorem 11, that the optimal entry probability, the bid bound and the shortlist size all fall in the entry cost, is the design-side cousin of P8.

Fu–Lu. $M \geq 3$ identical, sequential entry with full observation of current participants, organiser chooses prize and a per-entrant fee or subsidy. Their equation (7), total effort equals budget minus N times the entry cost, is again the P7 accounting identity, and is what drives their Theorem 1 that the optimal contest attracts exactly two entrants. Their conclusion lists different types of contestants as future research. So the entire design literature on endogenous entry — Fu–Lu 2010, Fu–Jiao–Lu 2015 — is homogeneous-agent by construction.

Morgan–Orzen–Sefton, and what the experiment did not vary. The theory section gives $n^* = \lfloor \sqrt{P/\bar{F}} \rfloor$ for identical agents with equal endowment w , outside option F and prize P . The experiment holds $w = 100$ for every subject by design. Predicted entry was 2 (small prize) and 4 (large prize); observed was 2.5 and 3.7 in the second half (rounds 26–50; 2.7 and 3.6 in rounds 1–25). The *signs differ*: the small prize shows *excess* entry (2.5 against 2, significant in both halves, $p = 0.004$ and 0.012) while the large prize shows a *shortfall* (3.7 against 4, differing from Nash only at the 6% and 12.5% levels). Both deviations run against what within-contest behaviour would predict — with negative returns to two-person competition one expects *fewer* than two, and with under-investment one expects *more* than four — and the paper says of each case “we observe precisely the opposite”. Citing this as “observed entry exceeds prediction” would be wrong for half the experiment. Earnings were *not* equalised across the contest and the outside option in either treatment, and a self-selection-by-risk-attitude explanation is rejected by comparing three-player contests across treatments (small-prize triples over-invest and earn less than the outside option, large-prize triples under-invest and earn more; the earnings difference is significant at 1%). The same controlled comparison also defeats a “contest premium” explanation, so two behavioural accounts are ruled out, not one. Two things follow. First, the experimental literature on contest entry has not tested resource sorting, because endowments were held equal; a Morgan–Orzen–Sefton design with heterogeneous endowments is the natural test of P8–P9. Second, their concluding observation that the marginal entrant’s power is blunted because inframarginal rent-seeking moves the marginal calculus has no analogue in the binary model, where inframarginal behaviour is fixed at “invest”; that is why the cutoff is clean here and noisy there.

Moreno–Wooders, read in full, and the relabelling objection answered. Moreno and Wooders (2011) has now been read in its published form [F]. N risk-neutral buyers in a standard independent-private-value auction; each buyer’s entry cost Z_i is an independent draw from H on $[\underline{c}, \bar{c}]$ and is *privately* observed before entry; the value is learned after entry; the seller may set a screening value v and an admission fee ϕ . Their Proposition 2 gives a unique symmetric equilibrium in threshold form: enter iff $z < t^*(v, \phi)$, where t^* solves $U(v, H(t)) = t + \phi$ and is decreasing in both instruments. The number of bidders is $\text{Binomial}(N, H(t^*))$. Proposition 3: $v = \phi = 0$ maximises social surplus, because the marginal type’s private gain equals its social contribution. Proposition 4: inframarginal entrants keep information rents, so the seller no longer captures the whole surplus. Propositions 5–8: the revenue-maximising screening value is positive but below the fixed- n level; with an admission fee available, screen by entry cost rather than by value; an entry cap plus fee does better still. Proposition 9: heterogeneity is irrelevant

asymptotically — only the lower bound $\underline{c}$ matters. Section 6: revenue may rise *or* fall in N , unlike Levin–Smith. There are no comparative statics in the distribution H anywhere in the paper.

The relabelling objection — is $\kappa \circ \Lambda^{-1}$ substantively different from a primitive cost distribution? — can now be answered on four points, and the first exposes a gap in `PROOFS.tex`.

Information. Moreno–Wooders’ costs are private, so the equilibrium is a symmetric Bayesian threshold and only the *distribution* of the entrant count is pinned. `PROOFS.tex` writes the entry condition as $\kappa(w_{(j)}) \leq \Delta(j-1)$ in terms of the order statistics of the wealth profile, which presupposes that the profile is common knowledge; that is what makes k^* deterministic and pins the identities of the entrants. This assumption is nowhere stated in `PROOFS.tex`. It must be, because it is the first structural difference from Moreno–Wooders and the reason P5 delivers an exact count where they deliver a binomial.

Primitive. Their H is a primitive and they never vary it. Ours is induced from a resource distribution through a known concave map, so a mean-preserving spread of Λ is a specific, non-arbitrary deformation of the induced cost distribution. P9 has no counterpart in Moreno–Wooders because there is no distributional comparative static there to have a counterpart to.

Designer. Theirs is mechanism design with three instruments; ours has none.

Objective. Their Proposition 3 is the independent-private-value logic where the marginal entrant’s private gain equals the social gain. For a fixed-prize winner-take-all contest the common-value logic of Levin and Smith (1994)’s Proposition 3 applies instead, and free entry is excessive. Their Proposition 4 carries over directly: entrants with $w > \bar{w}$ keep surplus.

So the threshold-in-cost-space is theirs; the count-pinning under complete information, the induced-cost structure, and the distributional comparative static are not. The objection is answered — but only once the information assumption is written down.

Still at abstract level. Kaplan and Sela (2010) give a rationale for entry fees in an all-pay auction with privately known entry costs. [C] Ginzburg (2021) has a continuum of contestants and a profit-maximising entry price. [A] Grossmann and Dietl (2015) has high- and low-talent contestants choosing between entry and heterogeneous outside options. [A] Samuelson (1985) is the interim-entry-cost variant (cost incurred after learning one’s value) that both Levin–Smith and Moreno–Wooders position against. [C]

5 Budget and liquidity constraints

Che and Gale (1997) has now been read in full [F], in its published form. The setting is $N \geq 2$ risk-neutral bidders with common value v , publicly known wealths $w_1 > \dots > w_N$, and a hard cap: a bidder cannot spend more than his wealth (no borrowing in §3; credit at rate r in §4). Utility is linear; wealth enters only as the cap. There is no entry cost and no participation decision.

Their Proposition 1 (all-pay auction) gives expected total expenditure $\min\{v, w_2\}$ with only the two wealthiest active — the rich preempt the poor. Their Lemmas 1–3 (lottery) give a unique equilibrium with a critical bidder i^* : those with wealth above a common bid b^* all bid b^* , those below bid their wealth, and everyone participates. Proposition 3: the lottery dissipates more than the all-pay auction iff v exceeds a threshold above w_2 . Adding low-wealth entrants has no effect in the all-pay auction and always raises lottery expenditure.

This reading confirms first-hand what §2 previously took from Schroyen–Treich’s description: a budget constraint is a kink in an otherwise linear payoff at w_i , and nothing else. The en-

try_contest model has no kink and no cap; $\kappa(w)$ is smooth and the sorting comes from curvature everywhere. The critical-bidder structure in their lottery is the hard-constraint cousin of the κ threshold, but it operates on the intensive margin (bid size) while ours operates on the extensive margin (enter or not), and in their lottery participation is never a choice. Their all-pay cap $\min\{v, w_2\}$ is wealth-driven and depends on the second-highest endowment; P7's cap V/κ is prize-driven and independent of the distribution. Their observation that adding low-wealth agents leaves all-pay expenditure unchanged rhymes with anonymity plus P7 — agents who would not enter do not move k^* — and is worth a sentence.

Grossmann and Dietl (2012) consider two bidders with asymmetric valuations under imperfect capital markets and find aggregate investment lower when at least one is constrained. [A] Grossmann and Hottiger (2020) and Che and Gale (1998) continue the line. [C]

6 Risk aversion and concavity in contests, read in full

Cornes and Hartley (2012) has been read in full as Manchester Economics Discussion Paper EDP-0806 (May 2008), the working version of the *Economic Theory* article. [F, WP version.] The setting is $n \geq 2$ contestants with logistic CSF and concave production, an indivisible common rent R , continuous expenditure, concave utility u_i , complete information; no entry cost.

The multiplicity example is now concrete. Their Example 1 has ten identical contestants with $u(c) = c - 0.45c^2$ — increasing absolute risk aversion — and possesses both the symmetric equilibrium with each spending 0.0563 and an asymmetric one in which three contestants spend 0.184 and seven spend zero; the active three earn more than in the symmetric equilibrium. Uniqueness (Theorem 4.2) needs *regularity*, and their sufficient condition (Lemma 4.2) is a curvature bound $2u'(R - x) \geq u'(-x)$ on $(0, R)$; it holds for CRRA when initial wealth $I_i \geq R(1 - 2^{-1/\gamma})^{-1}$ (Corollary 4.1), for small rents (Corollary 4.2), for prudent contestants with absolute risk aversion at most $1/(2R)$ (Corollary 4.3), and for CARA. Whether non-increasing absolute risk aversion suffices is left open.

The contrast with P5 is now exact. Their asymmetric equilibrium is a coordination among *identical* agents on who is active, sustainable because the continuous aggregative best response admits several fixed points under IARA. P5 has agents ordered by $\kappa(w)$, so who is active is pinned by the order and there is nothing to coordinate on; P5 needs only that κ is decreasing, with no curvature bound of the Lemma 4.2 type. That is the reason the binary-entry structure escapes their multiplicity, and it should be stated in one sentence with their Example 1 as the foil.

Two further results are directly relevant. Their dropout value $\bar{Y}_i = f'_i(0)[u_i(R) - u_i(0)]/u'_i(0)$ (Lemma 3.2) makes a contestant inactive whenever aggregate lobbying exceeds it — a curvature threshold on the *aggregate*, where $\kappa(w)$ is a curvature threshold on *own* resources. And Proposition 6.2 is a selection result: in large asymmetric contests only the types with the highest dropout value stay active, so competition drives the more risk-averse out and the limiting dissipation ratio $H[u] = [u(R) - u(0)]/(Ru'(0))$ is that of the least risk-averse type. That is selection by risk attitude across heterogeneous u_i with common wealth; ours is selection by resource level with common u . The two are complementary and should be cited together. Their CRRA specification $u(c) = (I_i + c)^{1-\gamma}/(1-\gamma)$ has initial wealth as an additive shift in a static game, the same one-period object as in Schroyen–Treich.

Treich (2010) has been read as LERNA working paper 09-013 of 17 February 2009 [F, WP version]. $n \geq 2$ identical agents with initial wealth w , concave u , rent b , payoff $p_i u(w + b - x_i) + (1 - p_i)u(w - x_i)$, general CSF. Proposition 2: prudence ($u''' \geq 0$) is sufficient for risk aversion

to lower effort relative to risk neutrality, for any n and any regular CSF; Corollary 1: for $n = 2$ it is necessary and sufficient, and quadratic u gives no effect at all. Proposition 1: DARA plus a small rent gives a unique symmetric equilibrium. The paper’s closing sentence lists entry as future work. Two things follow. The risk-aversion strand needs u''' to sign anything, whereas P8 and P9-gen need only u'' ; that is a concrete statement of what “no higher-order restrictions” buys. And wealth here is again a one-period additive shift. Skaperdas and Gan (1995) is the antecedent; Cornes and Hartley (2005) is the general-technology companion; Cornes and Hartley (2003) establishes regularity under CARA. [C]

7 Competitor numbers and the shape of noise, read in full

Ryvkin and Drugov (2020) has been read in its published *Theoretical Economics* form [F], and Drugov and Ryvkin (2020a) as the working paper of 13 March 2019 [F, WP version].

The setting is symmetric risk-neutral players, output $y_i = e_i + X_i$ with i.i.d. additive noise, a winner-take-all prize normalised to one, strictly convex cost, and a number of players K that may be deterministic or stochastic; heterogeneity is assumed away explicitly (their footnote 3). There is no entry margin. The two objects that drive everything are

$$b_k = \mathbb{E}[f(X_{(k-1:k-1)})], \quad E_k^* = k c'^{-1}(b_k) = \mathbb{E}[h(X_{(k-1:k)})] \text{ under quadratic cost,}$$

the density of noise at the largest of the other $k - 1$ draws, and the hazard rate at the second-largest of k . Proposition 1 and Corollary 1: a unimodal noise density is inherited by individual effort as a function of k , by a Karlin variation-diminishing argument on $F^{k-1}(1 - F)$. Corollary 4: monotone density gives monotone effort, constant iff uniform. Proposition 2: symmetric unimodal density gives $e_2^* = e_3^*$ and decreasing thereafter. Proposition 3: IFR noise with cost more convex than quadratic gives aggregate effort increasing in K ; DFR noise with cost less convex than quadratic gives aggregate effort decreasing. Proposition 5: for a mean-preserving increase in uncertainty about K , log-concave noise lowers aggregate effort and log-convex raises it. Proposition 7 (Appendix A.1) gives existence and uniqueness of the symmetric pure-strategy equilibrium under a lower bound on c'' and bounds on f and f' .

The relationship to P-MU can now be stated at the level of objects rather than by analogy. P-MU’s $\Delta(0)$ is the win-probability gain to the first investor, a functional of the induced score distributions; its dependence on Q runs through the same order-statistic weighting that gives b_k its dependence on k . Proposition 1’s inheritance of unimodality is the structural template for P-MU’s hump-shaped weighting, and the sentence on their p. 1589 — that no prediction about the effect of competitor numbers survives for every noise distribution — is exactly the frame in which R2’s Q -conjecture should be refuted. The next step is mechanical: write $\Delta(0)$ in the form $\int \phi dF(\cdot)$ and check whether their log-supermodularity condition on $-\tilde{G}_\theta$ is what P-MU’s weighting needs. Their footnote 29 records that Drugov and Ryvkin (2020b) shows the dispersive order is necessary and sufficient for ranking equilibrium effort across noise distributions — the dispersive order again, now in the contest branch as well as the status branch.

The 2019 working paper is the prize-allocation companion. With prizes summing to one and monotone in rank, the optimum is a two-prize schedule with the top r^* ranks paid equally (Proposition 1), where r^* maximises $B_{r,n}/r = \frac{1}{n} \mathbb{E}[h(Z_{(n-r:n)})]$; IFR gives winner-take-all, DFR gives reward-all-but-last (Proposition 2); the convex transform order gives monotone comparative statics in skewness (Proposition 4); log-concave noise makes effort Schur-convex in the prize vector (Proposition 7). Proposition 8 applies this to status tournaments in the sense of Moldovanu et al. (2007): with IFR noise the top status category is a singleton, with DFR it contains $n - 1$. This is not P-MU territory — the entry_contest model has one prize — but it is the status-tournament link back to the abandoned paper’s motivation, and a referee from that literature will expect it cited. Lim and Matros (2009) is the binomial- K special case. [C]

8 Assignment models and the pivot rule, read in full

Two papers previously listed as “assignment-model comparators” turn out to carry the closest published precedent for P9-gen’s pivot rule. Costrell and Loury (2004) has been read as the draft of 11 December 2003 [**F**, **draft version**] and Suen (2007) in its published form [**F**].

Costrell–Loury and the single-pivot precedent. In their two-job model (§2), jobs are filled in proportions θ and $1 - \theta$, and the entire wage schedule is determined by one sufficient statistic: $\hat{\mu} = F^{-1}(\theta)$, the ability of the *marginal* worker between jobs. Any shift that raises $\hat{\mu}$ raises low-ability wages and lowers high-ability wages. Under a mean-preserving spread on fixed support, they state that the effect depends on whether the marginal worker at quantile θ lies in the upper or lower tail: if θ is high the spread raises $\hat{\mu}$ and narrows the wage distribution, if θ is low it lowers $\hat{\mu}$ and widens it. That is a single-pivot sign rule located at the marginal agent’s quantile relative to the spread’s crossing point — the same logic as P9-gen, where the sign of dk^* flips according to whether $w_{(k^*)}$ lies above or below x_0 . Their object is a wage schedule through $\hat{\mu}$; ours is an entry count through $w_{(k^*)}$. P9-gen should be presented as the entry-count analogue of the Costrell–Loury two-job comparative static. Their continuum result (Proposition 6) then shows what governs the direction when there is no single marginal agent: with G riskier than F on $[0, 1]$, concave β lowers $w(0)$ and widens the span, convex β lowers $w(1)$ and narrows it. The curvature of the payoff map decides. For us that map is κ .

Three further results are tools. Their Lemma 1 (in the proof of Proposition 5) shows that G riskier than F on fixed support implies the quantile functions reverse the order, F^{-1} riskier than G^{-1} ; their Proposition 5 signs an aggregate (output) under a general spread with multiple crossings by integrating $\Gamma(\rho) = \int_{\rho}^1 [G^{-1} - F^{-1}]$ against $d\beta$; and their Theorem 1 shows that a Blackwell-more-informative signal induces a second-order-riskier distribution of posterior means, so any mean-preserving-spread result has an information reading. Their Proposition 10 (Cobb–Douglas crowding) removes the curvature dependence altogether: a spread always narrows the span. The draft is not the published article; the published version should be checked for these proposition numbers.

Suen and the general-spread technique. One-to-one transferable-utility matching with $Q = \theta(x)\phi(y)$, positive assortative matching, and matching function $\mu = G^{-1} \circ F$, which they note can itself be read as a distribution function. Proposition 2: if G_1 is more dispersed than G_0 (same mean, second-order), then total earnings on the other side fall, under concavity of θ, ϕ and log-concavity of $1 - F$. The proof handles a *general* Rothschild–Stiglitz spread with an arbitrary finite number of crossings $k_0 < k_1 < \dots < k_{2n}$ of the quantile functions: the quantile difference alternates in sign, each partial integral is signed by second-order dominance, concavity controls the recombination, and the result is integrated by parts against $\rho = (1 - F)/f$. Proposition 3 and its corollaries give conditions under which every agent loses. Suen’s μ is cross-side (the other market’s distribution moves) whereas our $T = \Lambda_{\beta}^{-1} \circ \Lambda_{\alpha}$ is same-side (before and after), but the map class is identical.

Consequence for open item 2. Two published sources sign an aggregate under a general mean-preserving spread, by the same first move but under *different* hypotheses — and they are not independent: Suen’s proof imports the quantile-reversal step from Costrell–Loury by citation (“see, for example, [2]”), and his footnote 1 engages their rank-rescaling device directly. One lineage, not two convergent confirmations.

Both begin by decomposing the quantile difference at the crossing points. They then diverge:

- Costrell and Loury (2004)’s Proposition 5 integrates $\Gamma(\rho) = \int_{\rho}^1 [G^{-1} - F^{-1}]$ against $d\beta$ and needs only that β is **non-decreasing**. It requires no concavity and no hazard-rate

condition; the strings “log-concav” and “hazard” do not occur anywhere in that paper. Concavity of β belongs to their Proposition 6, a different result about wage *inequality*.

- Suen (2007)’s Proposition 2 integrates by parts against $\rho = (1 - F)/f$ and needs concavity of θ, ϕ and log-concavity of $1 - F$ — the latter being exactly equivalent to $\rho' \leq 0$, since $d^2 \log(1 - F)/dx^2 = \rho'/\rho^2$.

The weaker hypothesis is Costrell–Loury’s, so the **monotone-weight route should be tried first**, not log-concavity.

Three obstacles stand between this and a result, none of them previously recorded here.

1. **Aggregate versus count.** Both sources sign a smooth integral (Q, W) . k^* is an integer count defined by a threshold crossing. Neither source signs a count, and this is the substantive gap.
2. **The rescaling trap.** The obvious route writes k^* as a functional of $\kappa \circ \Lambda^{-1}$ — a rank-space rescaling, which is precisely what Suen’s footnote 1 declines to do: “unless the density function g is increasing, concavity of ϕ does not imply concavity of $\hat{\phi}$ ”. Curvature does not survive the change of variable. Since the dispersive-order machinery of §3 that motivates $T = \Lambda_{\beta}^{-1} \circ \Lambda_{\alpha}$ is rank-space machinery, this sits on the main road, not a side path. Either establish curvature in the rescaled variable directly, or accept an increasing-density restriction on Λ that the model does not currently make.
3. **Cross-side versus same-side.** Suen compares two populations across a market; we compare one population before and after a spread. The map class is identical; the object compared is not.

This remains a concrete, checkable lead. It has not been tried and should not be claimed until it has.

9 Positional competition and education

Chung and Lee (2017) is the nearest thing to the abandoned paper’s framing: inequalities in wages, productivity and education affordability determining educational effort in competition for premier jobs, in a Tullock framework with endogenous returns, finding that competition for better-paying jobs can produce socially inefficient excessive effort and that a widening wage gap intensifies the competition. [A] Two players, continuous effort.

10 Foundations for a rank-allocated prize, read in full

Three papers previously cited secondhand have now been read in full, and they do a specific job: they justify the premise that `PROOFS.tex` currently asserts, that a fixed prize V is allocated by rank rather than by price.

Cole–Mailath–Postlewaite 1992. Published form [F]. Multigenerational; a continuum of men with capital and women with a non-traded endowment; joint consumption; CRRA. Status is defined as a ranking device that determines how one fares in the non-market sector. Two social norms: wealth-is-status and aristocratic (inherited). Proposition 1: a wealth-is-status equilibrium exists (Mas-Colell’s continuum-game theorem). Property 1: in equilibrium no family’s relative rank ever changes, even though everyone competes. Property 2: everyone weakly over-saves relative to the no-matching benchmark, except the bottom agent, who is undistorted. Proposition 2: an aristocratic equilibrium exists if the capital distribution is sufficiently spread out at the lower tail. Their footnote 20 says a market in which relative position determines the

house one buys *cannot* generate the multiplicity they are after — the prizes must be non-market. Their §IV.A two-period example states that the agent in the economy with the more compact income distribution tends to have the higher savings rate, and §V.D that once the distribution is dispersed enough the incentive disappears. That is the 1992 origin of the “standard result” R2 invoked, on the intensive margin.

Cole–Mailath–Postlewaite 1995. CARESS Working Paper 95-14, 1 August 1995, marked forthcoming in the Minneapolis Fed *Quarterly Review* [F, WP version]. The working-paper title has “Concern” singular; Hopkins–Kornienko’s 2010 reference list has “Concerns” plural at QR 19(3), 12–21. The published title has not been confirmed and the bib entry carries the working-paper form. The concern for relative standing is *instrumental*: rank matters only because a non-market decision depends on it. §2 is a complete-information effort model with the matching function $m(y)$ equal to the distribution function of output, so m' is large when the distribution is tight and competition is intense (§2.1, §2.3). §3 is the incomplete-information version: wealth is unobservable, so the rich signal by destroying wealth, and the fraction destroyed rises with rank. Footnote 8: it is competition from *below* that distorts effort — truncate the population from the top and nothing changes; truncate from the bottom and the new lowest agent becomes undistorted. §4: when prizes are allocated by rank rather than sold, the welfare theorems do not apply.

What these two give the entry_contest model. Finding 5 asks why threshold entry into a rank-allocated prize is not already covered. The answer to the “rank-allocated prize” half is that Cole–Mailath–Postlewaite is where that premise is justified and where its consequences (no welfare theorems, non-market allocation, instrumental status) are worked out. PROOFS.tex should cite them at the point where V is introduced and say that V is a prize in their sense — not sold, allocated by rank. Their “competition from below” is the same structural fact as the anonymity proposition and the ordering by κ : only agents who could outrank you matter. Their Property 1 is consistent with the model here — entry does not change the wealth rank; it is the score, not the rank, that the prize follows. And their multiplicity is *across* norms, not within one, so it is not in tension with P5.

Becker–Murphy–Werning. Becker et al. (2005), draft revised December 2003 [F, draft version]. Utility $u(c, s)$ with the crucial assumption $u_{cs} > 0$: status raises the marginal utility of consumption. A fixed supply of status positions, sold in a hedonic market (or, equivalently by their Proposition 1, assigned by rank in consumption of a fixed-supply, intrinsically worthless social good). Stage one: fair lotteries over income. Proposition 2: if the constant-marginal-utility allocation is a mean-preserving spread of the initial income distribution, it is the competitive lottery equilibrium — a minimum equilibrium inequality that any more compact initial distribution is spread out to reach. Proposition 3: with a status market this coincides with the utilitarian planner’s allocation, so the planner *increases* inequality. §7: when status attaches to income *rank* rather than being purchased, rank lotteries impose real externalities (one’s win lowers others’ rank) and the equilibrium no longer coincides with the planner’s. This is not a contest paper; the mechanism is orthogonal to κ , the mean-preserving spread is an *output* of equilibrium rather than an input, and $u_{cs} > 0$ is the structural opposite of our separable V . It matters for one thing only: the welfare section (R2 Comment 3). Their §7 states in print that rank-allocated status carries real externalities that purchased status does not, which is the zero-sum, business-stealing frame the welfare benchmark needs, and their Proposition 3 is a warning that a planner may *want* inequality.

Chung and Lee (2017), Cole et al. (1998), Postlewaite (1998), Frank (1985), Fernández and Galí (1999) and Bilancini and Boncinelli (2012) remain at abstract or citing-paper level.

11 Refuted conjectures in context

Referee 2 conjectured that for $\mu > 0$, higher Q shrinks the pie among investors so fewer invest. PROOFS.tex §R2 refutes this: the sign depends on μ , and the crossover μ^* is distribution-dependent. Placed against Ryvkin and Drugov (2020), this refutation is the expected outcome rather than a surprise, and framing it that way is both more honest and more persuasive than presenting it as an isolated counterexample.

12 Background and toolkit

Lazear–Rosen, read in full, and a precedent that was hiding in plain sight. Lazear and Rosen (1981), published form [F]. The risk-neutral two-player tournament: $C'(\mu) = (W_1 - W_2)g(0)$, efficient, with the prizes in equation (10) equal to expected product plus or minus $V/2g(0)$, which they read as an *entrance fee or bond*: each player posts a bond and takes a fair winner-take-all gamble over the pool. Footnote 2: a pure-strategy equilibrium exists only when the noise variance is large enough — the ancestor of Ryvkin–Drugov’s Proposition 7. §III adds risk aversion, and the subsection titled “Income Distributions” contains the result that matters here. With a common utility function and different *endowed* incomes y_0 , decreasing absolute risk aversion makes the rich prefer the tournament and the poor prefer the piece rate, so agents *self-select into the contest by wealth*, and the resulting income distribution is positively skewed. That is the first appearance, in 1981, of “who enters a contest is determined by wealth”. The channel is risk attitude — DARA, so the wealthy find the gamble cheaper — not the utility cost of a fixed fee, and it needs u''' where κ needs only u'' . But it is a precedent for the mechanism and it should be cited as one, with the channel distinction stated. §IV (heterogeneous ability, asymmetric information) gives adverse selection into the top league and a competitive handicap $h^* = \Delta\mu/2$; the handicap algebra is the tool for R2’s open question about an equilibrium in which the incumbent abstains and a challenger invests.

Moldovanu–Sela and Hoppe–Moldovanu–Sela, read in full. Moldovanu and Sela (2001) as the draft of 5 June 2000 [F, draft version]: $k \geq p$ contestants with *private ability* entering a separable cost $c_i\gamma(x)$, all-pay, perfectly discriminating; a single first prize is optimal under linear or concave cost (Propositions 3.2, 4.2) and two prizes can be optimal under convex cost (Proposition 4.3). Their §5 entry fee produces a threshold in *ability* space — a third threshold variant, alongside Moreno–Wooders’ cost space and our resource space. Their footnote 12 warns that independence of types is problematic with endogenous entry, which applies here too: the entrant pool is the top k^* by w , not an i.i.d. sample. Hoppe et al. (2009), published form [F]: finite two-sided matching on costly signals with private attributes, built on Barlow and Proschan (1966)’s normalised-spacings results. Proposition 4: entry on the long side raises own-side signalling for every F , and moves the other side’s signalling up or down according to whether F is DFR or IFR; Example 1 shows welfare falling in n up to $n = 8$ in a small market and rising thereafter. Proposition 5: the relatively *more homogeneous* side signals more, because it perceives a more dispersed prize structure. That is the perfectly-discriminating, private-ability counterpart of the competitor-number comparative static: with no noise, entry always raises own-side effort; P-MU says that with noise the sign at the margin is distribution-dependent even for symmetric agents. Their IFR/DFR spacings machinery and Ryvkin–Drugov’s hazard-rate machinery are the same Barlow–Proschan lineage.

Surveys and standard references. Konrad (2009) and Corchón (2007) are the two surveys a referee will expect to see acknowledged; the 2010 Hopkins–Kornienko text cites Konrad for the point that greater heterogeneity among competitors and a smaller spread of prizes both reduce equilibrium effort. [F, as characterised there.] Moldovanu and Sela (2006) is the

contest-architecture companion and Schotter and Weigelt (1992) the experimental treatment of asymmetric tournaments. [C]

Two references carry technical weight rather than positioning weight. Shaked and Shanthikumar (2007) is the source for the dispersive and star orders and is the right citation for the pivot-spread class discussed in §3. Rothschild and Stiglitz (1970) is the definition open item 2 of `PROOFS.tex` appeals to when it observes that a general mean-preserving spread admits only a mean-preserving coupling $w' = w + \varepsilon$ with $\mathbb{E}[\varepsilon \mid w] = 0$, which need not be monotone. Citing both, and stating that the pivot-spread hypothesis is the dispersive-order restriction rather than an arbitrary one, converts that open item from an admitted weakness into a deliberate and named choice of stochastic order.

13 Verdict on novelty

The mechanism alone does not carry a paper, and after the full read of Schroyen and Treich (2016) it carries even less than the previous draft of this document supposed. “Resource-sorted entry via concavity” is their privilege contest plus a Moreno and Wooders (2011) threshold on the entry structure, and the resource-label distinction does not separate the two (§2). Nor is a mean-preserving spread in the resource distribution new in itself: their Theorem 3 already signs one inside a privilege contest.

Two further sharpenings from the full reads. Lazear and Rosen (1981)’s “Income Distributions” subsection (within §III) already has agents self-selecting into a tournament by endowed wealth, through DARA; the mechanism has a 1981 precedent and the contribution sentence must acknowledge it. Two qualifications, both verified, bound how much it concedes.

First, *their claim is an example and they say so*. The subsection opens “While it is not possible to make a general argument based on an example, table 1 suggests. . .”, and rests on one utility ($U = ay^a$), quadratic costs, normal errors and two endowment levels. The honest acknowledgement is that Lazear and Rosen *suggest* the sorting on an example under DARA, while the present model *derives* it under concavity alone, distribution-free.

Second, the channel difference (u'' here, u''' there) is substantive rather than verbal, and separates decisively. Our $\kappa(w) = u(w) - u(w - c)$ is strictly decreasing whenever $u'' < 0$, with no condition on u''' ; DARA needs $A'(w) < 0$, whose numerator is $(u'')^2 - u'''u'$, so it *requires* $u''' > 0$. Quadratic utility witnesses the gap: it is concave, has $u''' = 0$, and is *increasing* absolute risk aversion — our sorting operates on it, theirs cannot.

A third difference, which the channel formulation understates, is the *object of choice*: theirs is a choice between two compensation schemes that both pay, so the sorting is tolerance for the binomial spread of a lottery; ours is whether to pay a fixed fee at all, so the sorting is affordability in utility units. That is different economics, not merely a different derivative, and it is the more robust half of the answer.

And Moreno and Wooders (2011) (§4) shows that what separates P5 from a private-cost threshold is the complete-information assumption on the wealth profile — which `PROOFS.tex` uses but does not state.

What is not in any source located here is the combination of three results that require the Q -player binary-entry setting jointly:

1. **P7**, the uniform cap $k^* \leq V/\kappa$ holding for every Q with no limit taken. The accounting logic behind it — entrants times entry cost cannot exceed prize mass — is already in Fu et al. (2015), equation (2) *together with their Definition 2* (they never display the count bound itself; see §4), and Fu and Lu (2010), equation (7), for identical agents with mixed

or sequential entry. Fu–Lu’s is the sharper case, (7) being an equality rather than an inequality — but their N is the count in an *optimally designed* contest, so “under fixed rules with no designer” is doing real work. P7’s content is the deterministic heterogeneous version in utility units, bounding a pure-strategy count rather than an expected one, under fixed rules with no designer. It should be presented as that, not as a new inequality.

2. **P9 and P9-gen**, the pivot rule locating the sign reversal of the inequality effect on aggregate entry at $w_{(k^*)}$ versus x_0 . The novelty is not that a mean-preserving spread is signed — Schroyen and Treich (2016) do that — but that it is signed *globally and without higher-order restrictions on u* , where the intensive-margin result is local and needs the CSF decisiveness parameter and the third derivative. The literature supplies the stochastic order (dispersive) but not the entry-count application. The closest structural precedent for the pivot rule itself is Costrell and Loury (2004)’s two-job model (§8), where the sign of a spread’s effect flips according to whether the marginal worker’s quantile lies above or below the crossing; P9-gen is its entry-count analogue and should cite it as such.
3. **P-MU**, the exact condition for the sign of the competitor-number effect in a binary-entry contest, as the discrete analogue of the Ryvkin and Drugov (2020) *density* result $b_k = \mathbb{E}[f(X_{(k-1:k-1)})]$ — *not* their hazard-rate result, which governs *aggregate* effort as $\mathbb{E}[h(X_{(k-1:k)})]$ and uses a different order statistic (the two densities stand in ratio $k(1-F)$). Since $\Delta(0)$ is an individual gain, b_k is the right comparator. Note that Schroyen and Treich (2016) list an arbitrary number of players as an extension they did not undertake, so the Q -dimension is unoccupied on both sides.

The comparison has now been computed rather than conjectured, and it is exact: P-MU’s weight is $W = G^{Q-1}(1-G)F$, whose hump-shaped factor $G^{Q-1}(1-G)$ peaks at $G = (Q-1)/Q$, while Ryvkin and Drugov’s log-supermodular kernel $-H_\theta = F^{k-1}(1-F)$ peaks at $F = (k-1)/k$. Under $G \leftrightarrow F$, $Q \leftrightarrow k$ these are the *same function*; our extra factor F is the incumbent’s CDF and carries no Q . So the hump-shape is not an artefact of our construction but the standard object of this literature. Log-supermodularity transfers (the cross-partial of $\log[z^{n-1}(1-z)]$ is $1/z > 0$), but the second hypothesis does not: Karlin’s argument needs the integrand in the role of u' to be single crossing $+-$, whereas ours is $F \cdot (f-g) = -F\varphi'$, which crosses $-+$. *No unimodality claim for $\Delta(0, Q)$ in Q is available*; the plausible target of a reworked argument is an interior *minimum*, which would sit consistently with R1–R2, and it has not been derived.

The contribution should be stated as those three theorems, not as the mechanism. That is a narrower claim than the rebuild handover assumed, and it survives.

14 What still needs doing

Read in full as of this version: Hopkins–Kornienko 2004, 2009, 2010; Schroyen–Treich (WP); Ryvkin–Drugov (TE, published); Drugov–Ryvkin (JET, WP); Fu–Jiao–Lu (published); Fu–Lu (WP); Cornes–Hartley (WP); Treich (WP); Levin–Smith (published); Morgan–Orzen–Sefton (published); Moreno–Wooders (published); Che–Gale 1997 (published); Suen (published); Costrell–Loury (draft); Cole–Mailath–Postlewaite 1992 (published) and 1995 (WP); Becker–Murphy–Werning (draft); Moldovanu–Sela 2001 (draft); Hoppe–Moldovanu–Sela (published); Lazear–Rosen (published). Every substantive claim in this document about what a paper does now rests on a full read, except where marked [A] or [C]. Where a working paper or draft was read, the published version has not been line-checked.

Outstanding, in descending order of consequence for the current claims:

1. State the complete-information assumption on the wealth profile in `PROOFS.tex` (§4).

Without it P5’s deterministic k^* is not distinguished from Moreno–Wooders’ binomial count.

2. Add Lazear and Rosen (1981) to the contribution paragraph as the precedent for wealth-based self-selection, with the u'' versus u''' distinction.
3. Cite Cole et al. (1992, 1995) at the point where V is introduced (§10).
4. Relate P-MU’s weighting W to Ryvkin and Drugov (2020)’s b_k and their log-supermodularity condition — a computation.
5. Try the Suen / Costrell–Loury general-spread technique on P9-gen (§8) — a proof attempt.
6. Published-version checks: Schroyen–Treich (*GEB*, Theorem 3 and the payoff specifications); Fu–Lu (*EI*, the title differs between MPRA and Fu–Jiao–Lu’s citation); Cole–Mailath–Postlewaite 1995 (*QR*, “Concern” or “Concerns”); Costrell–Loury, Becker–Murphy–Werning, Moldovanu–Sela, Cornes–Hartley, Treich, Drugov–Ryvkin (proposition numbering).
7. Grossmann and Dietl (2015) and Chung and Lee (2017), for the entry-condition comparison and the welfare template respectively — still at abstract level.

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